Formal Review of the Round-6 Consolidated Generated Split-Forest Proof

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Abstract

This memorandum reviews the attached manuscript `repaired_split_forest_no_automata_proof_v2_consolidated_round6` and its TeX source. The round-6 document is a clear improvement over the previous consolidated version. It compiles cleanly, it directly addresses the flat side-frontier problem by introducing recursive verticalization, it replaces per-triple mosaic coverage by an explicit abstract label function L_Q, it adds cross-pair universal safety, and it strengthens the request-cycle argument with half-open cycles and extended profiles. Nevertheless, I would not yet regard the decidability proof as closed. The most serious remaining issue is that the new abstract-position labelling mechanism appears to reintroduce the blocking/equality conflation it was designed to avoid: if all occurrences with the same abstract position symbol are labelled by $L_Q(\pi, \pi) = \mathsf{EQ}$, then distinct laps of a blocking cycle become semantically equal. This affects the basic one-state upward-chain certificate. Additional remaining issues concern the occurrence-local interpretation of equality ports, the precise stratum identity stored in the finite certificate, the recursive side-interface construction, universal safety across pumped cycles, and inconsistent side-width bounds. My verdict is therefore: substantial progress, but not yet a fully watertight decidability proof.

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1 Materials reviewed

I reviewed the following uploaded files:

- `repaired_split_forest_no_automata_proof_v2_consolidated_round6.pdf`;
- `repaired_split_forest_no_automata_proof_v2_consolidated_round6.tex`.

The TeX source compiles locally without fatal errors and produces a 39-page PDF. I did not receive a new verification archive with this round, so this review is a manuscript review rather than a rerun of an updated executable verification suite.

2 Executive verdict

The round-6 manuscript makes real progress. In particular, it no longer tries to defend the old flat side-frontier discipline. The new stratified side interface and recursive verticalization discipline (M6′) are the right kind of repair for side witnesses that are themselves PP/PPI-comparable. The manuscript also correctly identifies the previous inconsistency between per-position support closure and per-triple mosaic coverage, and it attempts to move the composition proof to the explicit global label function $L_{\mathcal{Q}}$. The request-cycle section is also more coherent than in the previous round.

However, several of the new repairs are not yet internally consistent. The most serious defect is independent of the support-mosaic details. The current combination of

$$q(u) \in \mathsf{Pos}_{\mathcal{Q}}, \qquad \mathsf{lab}(u,v) := L_{\mathcal{Q}}(q(u), q(v)), \qquad L_{\mathcal{Q}}(\pi,\pi) = \mathsf{EQ}$$

seems to force every two occurrences with the same finite abstract position symbol to be EQ-related. Since blocking cycles necessarily reuse finite abstract position symbols on fresh semantic occurrences, this collapses cyclic blocking into semantic equality. This directly conflicts with the manuscript's central slogan that blocking is not equality.

Thus my revised verdict is:

$$\boxed{\text{Round 6 is an important improvement, but the proof still does not stand as written.}}$$

The proof architecture remains credible, but the formal certificate semantics needs another occurrence-level repair.

3 Positive changes in round 6

3.1 Stratified side interfaces address the old flat-frontier problem

The earlier manuscript treated side/frontier positions too flatly. That lost satisfiable configurations in which two DR/PO side witnesses of the same source are themselves PP/PPI-comparable. The round-6 text now explicitly recognizes this and introduces a stratum forest inside a mosaic. This is the right conceptual repair: a pair of comparable side witnesses should be co-vertical inside a nested local stratum rather than rejected as an illegal sibling-frontier pair.

3.2 The proof no longer depends on per-triple mosaic coverage

The previous version weakened support closure to per-position support but later continued to appeal to per-triple mosaic coverage. The round-6 manuscript tries to repair this by using the declared global pair-label function

$$L_{\mathcal{Q}} : \mathsf{Pos}_{\mathcal{Q}}^2 \to \mathsf{B}$$

and enforcing the RCC5 composition table directly by validity clause (V11). This is a promising direction. If the abstract pair labels are defined over the correct kind of occurrence-pair shapes, then composition can indeed be checked finitely without requiring every triple to appear inside one mosaic.

3.3 Cross-pair universal safety is now explicit

The new validity clause (V15) is also the right sort of fix. RCC5 composition alone does not propagate DL types; if an overlap cross-label is declared to be R, then universal restrictions $\forall R.D$ at the source must be checked against the target type. Making this a finite validity condition is an improvement.

3.4 The request-cycle section is more precise

The switch to half-open cycle words $[i, j)$ and the use of extended profiles including pending requests are both improvements. This avoids one of the previous defects, where a repeated ordinary profile might not carry the same unfulfilled eventualities.

4 Main remaining blocker: abstract positions collapse blocking into equality

The central formal problem is the new definition of concrete labels through the finite abstract position map. The manuscript defines an abstract position symbol $q(u) \in \mathsf{Pos}_{\mathcal{Q}}$ for each concrete occurrence u and then defines

$$\mathsf{lab}(u, v) := L_{\mathcal{Q}}(q(u), q(v)).$$

It also requires

$$L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}.$$

This is harmless only if $q(u) = q(v)$ implies $u = v$. But in a finite regular certificate, this implication is false. Blocking deliberately creates infinitely many fresh semantic occurrences with the same finite profile, port, context, and stratum data.

4.1 Concrete failure on the upward-chain stress case

Consider again

$$C_{\uparrow} = \exists \mathsf{PP}.\top \ \sqcap \ \forall \mathsf{PP}.\exists \mathsf{PP}.\top.$$

The intended regular certificate is the one-state upward loop whose unfolding is

$$u_0 \mathsf{PP} u_1 \mathsf{PP} u_2 \mathsf{PP} \cdots,$$

with all u_i fresh occurrences of the same finite state/profile. In such a certificate, the occurrences u_i and u_j for $i \neq j$ will have the same abstract position symbol:

$$q(u_i) = q(u_j) = \pi.$$

The round-6 labelling rule then gives

$$\mathsf{lab}(u_i, u_j) = L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}.$$

This is impossible. The intended relation for $i < j$ is PP, and strong EQ semantics forbids two distinct occurrences from being EQ-related. If the quotient identifies them, the strict PP chain collapses into a PP-cycle. If the quotient does not identify them, the pre-quotient or quotient frame has EQ between distinct elements. Either way the current formal definitions do not represent the very blocking cycle they are meant to represent.

4.2 Why this is not just notation

This defect affects several later arguments:

- the proof of the support-closed patchwork theorem uses $\mathsf{lab}(u, v) = L_{\mathcal{Q}}(q(u), q(v))$ for arbitrary concrete pairs;
- Lemma 7.4/JEPD relies on EQ iff same quotient class;
- the typed-EQ quotient proof assumes blocking repetition is not semantic equality;
- V9 forbids equality between strict vertical comparables.

These cannot all be true if $L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}$ is applied to two distinct fresh occurrences with the same abstract symbol.

4.3 Required repair

The label function cannot be a plain function

$$L_{\mathcal{Q}} : \mathsf{Pos}_{\mathcal{Q}}^2 \to \mathsf{B}$$

with diagonal EQ unless abstract positions are occurrence-unique, which would make the certificate infinite. A finite certificate needs a label function over *abstract occurrence-pair shapes*, not just over pairs of position symbols. At minimum the pair shape must distinguish:

(D1) the literal identity pair (u, u), where the label is EQ;

(D2) two distinct occurrences with the same abstract position symbol but in different laps or different context instances;

(D3) vertical successor/predecessor copies across a blocking cycle;

(D4) side/mosaic-visible pairs;

(D5) explicit equality-port pairs.

One possible replacement is a finite function

$$L_{\mathcal{Q}}^{\text{pair}} : \mathsf{PairShape}_{\mathcal{Q}} \to \mathsf{B}$$

where $\mathsf{PairShape}_{\mathcal{Q}}$ contains an explicit identity flag and enough relative-copy information to distinguish same-symbol identity from same-symbol fresh copies. The validity clause (V11) would then quantify over compatible *triple shapes*, not over $\mathsf{Pos}^3_{\mathcal{Q}}$ alone.

5 Equality-port lifting remains ambiguous

A related issue occurs in the definition of the occurrence lift of $\equiv^Q_{\mathsf{EQ}}$. The manuscript says that $\equiv^Q_{\mathsf{EQ}}$ is the reflexive, symmetric, transitive closure of explicit port links, and then writes informally that

$$u \equiv_{\mathsf{EQ}} v \quad \text{iff} \quad \mathsf{Port}(u) \equiv^Q_{\mathsf{EQ}} \mathsf{Port}(v)$$

within the same generated context or transitively across explicit links. If this is read literally at the state-port level, then every occurrence with port (s, p) is equivalent to every other occurrence with port (s, p), because $\equiv^Q_{\mathsf{EQ}}$ is reflexive. Again, distinct laps of a blocking cycle collapse.

There is also tension with V9.b. The manuscript says that a finite cycle counts as strict reachability because each lap unfolds to a fresh occurrence. Thus a self-loop can make

$$(s, p)\ \mathsf{reach}_{\mathsf{PP}}\ (s, p).$$

But $\equiv^Q_{\mathsf{EQ}}$ is reflexive on (s, p), so V9.b, read literally at the finite state-port level, would reject the very self-loop needed for $C_\uparrow$. If V9.b is not meant to reject the reflexive port occurrence, then the text must say so and must define the occurrence-level lift explicitly.

Required repair. The finite port relation should be treated as a *template* for explicit local equality edges generated inside a context instance. Its lift should be:

$$u \equiv_{\mathsf{EQ}} v \quad \Longleftrightarrow \quad u = v \text{ or there is a path of instantiated explicit equality edges between } u \text{ and } v.$$

It should not identify all occurrences with the same finite port. V9.b should then be stated over instantiated occurrence-level equality edges, or over pair shapes that can actually occur in one context instance, not over reflexive state-port equality in the finite quotient.

6 Stratum identity is still not fully present in the finite abstraction

The abstract position definition still uses a stratum field

$$\zeta \in \{\mathsf{vert}, \mathsf{sib}\}.$$

But M6′ is no longer a binary vertical/sibling distinction. It uses a full stratum forest

$$\mathcal{V} = (\mathsf{Stratum}, \mathsf{Subsrc})$$

with nested strata. Whether two positions are co-vertical depends on the identity of the stratum containing them, not merely on whether each is marked "vertical". Two positions can both be vertical but in different strata; they are then cross-stratum and should not receive PP/PPI labels.

The introduction says the round-6 certificate grammar tracks the full stratum forest object $\mathcal{V}$, but the formal abstract-position definition still looks round-4-like. The permitted-pair condition says that co-verticality is "lifted via the certificate's stratum-membership data", but that data is not part of the abstract position symbol as written.

Required repair. Replace the binary field $\zeta \in \{\mathsf{vert}, \mathsf{sib}\}$ by a stratum identity or stratum-path field, for example

$$\zeta \in \mathsf{StratumId}(C),$$

where C is the context scheme. Then define co-verticality in Pos_Q by equality of an actual stratum identifier. Validity clauses (V2), (V11), (V12), and the enumeration grammar should all use the same stratum-forest object, not the old Vert notation or a binary vertical/sibling flag.

7 The recursive side-interface construction is improved but still underspecified

The new side-interface legality lemma is much closer to the needed statement. However, the construction of nested strata is still ambiguous for configurations with more than two comparable side positions.

The lemma says that for every side witness w whose original is PP/PPI- related to another side position w', a nested stratum V_w is introduced. But if one side position is comparable to several others, or if the side positions form a chain

$$w_0 \mathsf{PP} w_1 \mathsf{PP} w_2,$$

it is not clear whether there is one stratum per comparable pair, one stratum per source w, or a recursively nested chain of strata. The stratum forest axioms allow overlaps only at parent-source positions, so the construction must specify exactly how such chains are represented without creating illegal stratum intersections.

There is also a tension inside M6′. Clause (M6′.b) says that a containment label in a stratum must match the source/mate versus selected-witness structure. That appears to allow PP/PPI labels only between the stratum source (or its mates) and selected witnesses. But the vertical-realisation proof later has a subcase where both endpoints are selected containment witnesses of the same stratum and $L(p, q) \in \{\mathsf{PP}, \mathsf{PPI}\}$. As written, that subcase seems forbidden by M6′.b rather than explained by it.

Required repair. Each stratum should carry an explicit local containment order or local parent map, not merely a source and a set of selected witnesses. Then M6′.b can say: $L(p, q) \in \{\mathsf{PP}, \mathsf{PPI}\}$ iff p, q are comparable in the local stratum order, possibly modulo equality mates. The recursive side-interface lemma should construct this local order for arbitrary finite comparable side-witness networks and prove the stated modal-depth width recurrence.

8 Request cycles handle existentials better, but universal wrap-around still needs proof

The request-cycle lemma now handles transitive existential requests more carefully. However, pumping a half-open segment can create new future successors for a universal formula. If a node u_k in the cycle has $\forall R.D$, then in the pumped path it sees not only later nodes $u_{k+1}, \ldots, u_{j-1}$ from the same lap, but also the next lap $u_i, u_{i+1}, \ldots$. The original infinite path did not require $u_i, \ldots, u_{k-1}$ to satisfy D, because they were earlier than u_k.

The proof currently says that universal truth is preserved because the visible neighbourhood is the same as in the original path. That is not literally true under cyclic pumping: the wrap-around part of the next lap is newly visible as a strict future successor.

This may be repairable. The profile already gestures toward storing formulas true at all/some strict ancestors and descendants. But the request-cycle lemma must use that information explicitly. A valid cycle should satisfy a finite universal-cycle safety condition:

$$\forall R.D \in \lambda(u_k) \quad \Longrightarrow \quad D \in \lambda(u_m) \quad \text{for every later position } m \text{ in the cyclic order.}$$

For a half-open cycle $[i, j)$ this includes the wrap-around positions $i \leq m < k$ in the next lap. The completeness proof then needs to show that a cycle satisfying this stronger condition can always be selected from a satisfying ray, perhaps by choosing the lasso after all universal-suffix summaries have stabilized.

9 The side-width and enumeration bounds are internally inconsistent

The round-6 introduction and Lemma 4.5 say that the old cubic side-width bound has been replaced by a modal-depth recurrence

$$m_*(C_0) = m(\mathsf{md}(C_0)).$$

The enumeration lemma also says "No cubic side-width is used." But the later "Explicit component bounds" section still defines

$$m(n) = 1 + n^3 + 3n^2 + 3n = O(n^3)$$

as a cubic polynomial bound on $|\mathsf{Side}(u)|$ and on mosaic arity. The same section then defines polynomial bounds using this cubic $m(n)$.

This is not merely a cosmetic mismatch. The finite search bound is the final step of the decidability proof. The proof needs one consistent bound, with the same symbol used throughout. If the correct bound is the modal-depth recurrence, then the grammar and enumeration estimates should use $m_*(C_0)$ everywhere. If a cubic bound is actually sufficient, then Lemma 4.5 and the surrounding round-6 explanation should be changed accordingly.

10 The Renz–Nebel / patchwork layer should be simplified or restated

The manuscript defines semantics as strong abstract atomic RCC5 frames, not as topological or set-theoretic region models. For this abstract semantics, soundness only needs JEPD, inverse symmetry, and RCC5 composition closure. The new L_Q-direct proof already aims to establish exactly these abstract frame conditions.

The Renz–Nebel realisability theorem is therefore not obviously needed for the main decidability theorem as stated. Worse, the "Moreover" clause in the current restatement is too weak as written: saying there exists some $\ell \in \mathsf{comp}(\rho_1(u,v), \rho_2(v,w))$ does not itself define a globally consistent cross-label for all mixed triples. The later strong-overlap condition supplies actual cross-labels, so the theorem statement should not suggest that mere non-emptiness of a composition entry is enough.

Recommendation. Either remove the set-theoretic Renz–Nebel realisation layer from the main proof, or clearly mark it as optional background. For the abstract-frame result, the patchwork theorem can be replaced by the direct statement: if the certificate supplies a well-defined pair-shape labelling satisfying JEPD, inverse symmetry, and V11 on all compatible triple shapes, then the unfolded structure is an abstract RCC5 frame.

11 Recommended repairs before claiming decidability

I recommend the following concrete repair sequence.

(R1) Replace $L_Q : \mathsf{Pos}_Q^2 \to \mathsf{B}$ by a label function on finite abstract occurrence-pair shapes. Include an explicit identity flag so that EQ on the diagonal applies only to literal identity, not to two fresh blocked copies with the same finite position symbol.

(R2) Redefine the occurrence lift of equality ports. Equality links should be instantiated local edges inside generated context occurrences; reflexivity of a finite port should not identify all occurrences using that port.

(R3) Rewrite V9.b at the occurrence-template level. It should forbid explicit equality between two positions that can be strict-vertical comparable in the same unfolding instance, without rejecting self-loop blocking profiles by reflexivity.

(R4) Make stratum identity part of abstract positions or pair shapes. A binary vertical/sibling flag is not enough for recursive M6′.

(R5) Strengthen M6′ so each stratum has an explicit local containment order, and rewrite the side-interface construction for arbitrary finite comparable side-witness networks.

(R6) Add a universal-cycle safety lemma for pumped half-open cycles, or strengthen the extended profile so that request-fulfilled cycles are also universal-safe across wrap-around.

(R7) Normalize the side-width bound: use either the modal-depth recurrence $m_*(C_0)$ everywhere or prove the cubic bound everywhere.

(R8) Simplify the patchwork layer to the abstract-frame needs of the theorem, or restate the Renz–Nebel component precisely and mark which parts are optional.

12 Conclusion

Round 6 is genuine progress. The manuscript has moved from the old flat side-interface picture toward the right recursive, occurrence-based split-forest architecture. The major conceptual repairs are in place: witness-generated covers, split copies, typed equality, recursive verticalization, cross-pair universal safety, and request-fulfilled cycles.

But the proof still contains a serious formal defect: the finite abstract position labelling currently equates distinct blocked copies by assigning $L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}$ whenever their abstract position symbols coincide. This undermines the central blocking-not-equality principle and breaks the canonical one-state certificate for

$$\exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top.$$

Until the label function and equality-port lift are repaired at the occurrence-pair-shape level, I would not regard the decidability theorem as established.

The architecture is still promising, but the next manuscript should focus on this single foundational correction before further polishing the mosaic and verification layers.